

MOPTI, Mali

WMO: 612650

Lat: 14.513N

Lon: 4.080W

Elev: 276

StdP: 98.05

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 14.4	(c) 15.8	(d) -8.2	(e) 1.9	(f) 30.9	(g) -8.0	(h) 2.0	(i) 31.1	(j) 9.4	(k) 27.9	(l) 8.5	(m) 27.4	(n) 1.6	(o) 40	(p) 0.433

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 5	(b) 12.0	(c) 42.2	(d) 20.6	(e) 41.5	(f) 20.6	(g) 40.6	(h) 20.7	(i) 27.1	(j) 31.6	(k) 26.6	(l) 31.4	(m) 26.2	(n) 31.3	(o) 3.6	(p) 40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.0	22.1	29.1	25.5	21.4	28.9	25.0	20.8	28.8	87.1	31.5	85.1	31.5	83.4	31.3	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.4	(b) 8.1	(c) 7.1	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 9.4	(b) 8.1	(c) 7.1	(e) 6.5	(f) 28.1	(g) 0.6	(h) 0.9	(i) 6.0	(j) 28.7	(k) 5.7	(l) 29.3	(m) 5.3	(n) 29.7	(o) 4.9	(p) 30.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	29.4	23.8	27.0	30.7	33.5	34.6	32.9	29.9	28.1	28.9	30.1	28.0	24.9	(6)
(7)		DBStd	3.78	2.34	2.91	2.81	1.89	1.77	2.21	2.09	1.54	1.49	1.33	1.59	2.23	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	1	1	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	7068	427	477	641	705	762	686	617	562	567	622	540	462	(10)
(11)		CDD18.3	4027	168	244	383	455	503	436	359	303	317	364	290	204	(11)
(12)		CDH23.3	55505	1768	3016	5688	7410	8381	6791	4763	3387	3806	4863	3493	2139	(12)
(13)		CDH26.7	32962	787	1692	3641	5104	5922	4456	2542	1416	1739	2647	1973	1043	(13)
(14)	Wind	WSAvg	3.1	3.1	3.4	3.3	3.2	3.6	4.1	3.6	2.8	2.4	2.3	2.6	3.2	(14)
(15)	Precipitation	PrecAvg	494	1	0	1	5	16	65	137	172	84	20	1	1	(15)
(16)		PrecMax	850	31	0	14	43	68	211	212	271	228	62	9	1	(16)
(17)		PrecMin	264	0	0	0	0	0	14	69	71	17	0	0	1	(17)
(18)		PrecStd	134	6	0	3	10	18	49	35	54	47	21	2	0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.1	39.5	41.8	43.2	43.4	42.3	38.9	35.5	36.2	39.0	38.1	36.4	(19)
MCWB			16.5	17.6	18.5	19.6	21.3	22.2	24.1	25.4	24.8	19.6	18.3	17.2	(20)	
2%		DB	33.7	37.9	40.6	42.1	42.3	41.0	37.4	34.1	34.9	37.5	37.0	35.1	(21)	
		MCWB	15.6	17.2	18.5	19.6	21.2	23.4	24.3	25.5	25.6	20.6	17.8	16.5	(22)	
5%		DB	32.3	36.7	39.5	41.3	41.6	39.8	36.1	33.1	33.9	36.3	36.1	33.8	(23)	
		MCWB	14.9	16.8	18.3	19.4	21.4	23.7	24.3	25.5	25.7	21.9	17.7	16.1	(24)	
10%		DB	31.0	35.0	38.3	40.3	40.6	38.5	34.8	32.2	32.9	35.1	35.0	32.1	(25)	
		MCWB	14.4	16.0	17.7	19.4	21.6	23.6	24.4	25.4	25.6	23.0	17.6	15.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.4	20.2	22.9	24.8	25.7	26.3	26.8	27.2	27.7	27.7	24.7	20.2	(27)
MCD B			30.3	34.6	36.2	36.9	35.8	35.7	31.9	31.6	32.0	31.5	30.0	32.4	(28)	
2%		WB	16.9	18.8	21.7	24.0	25.1	25.7	26.2	26.6	27.1	26.7	23.4	18.7	(29)	
		MCD B	30.0	33.5	34.3	35.0	36.1	35.2	32.0	31.2	31.6	31.3	29.6	29.8	(30)	
5%		WB	16.0	17.9	20.5	23.3	24.6	25.2	25.8	26.2	26.7	26.3	21.8	17.6	(31)	
		MCD B	29.9	33.6	34.2	34.7	36.0	34.5	31.8	30.7	31.2	31.2	28.6	29.6	(32)	
10%		WB	15.1	16.9	19.2	22.6	24.1	24.8	25.4	25.9	26.3	25.8	20.4	16.6	(33)	
		MCD B	28.8	32.8	34.6	34.5	35.7	34.0	31.4	30.3	30.9	31.0	29.1	29.1	(34)	
(35)	Mean Daily Temperature Range	MDBR	14.5	14.6	14.3	13.3	12.0	10.4	8.9	7.6	8.2	10.7	14.5	14.8	(35)	
5% DB		MCD BR	16.3	16.2	15.3	14.7	13.2	11.7	10.2	8.9	9.3	12.9	15.8	16.2	(36)	
		MCWBR	5.7	5.5	5.0	5.0	4.2	3.2	2.4	2.8	3.4	4.4	5.1	5.5	(37)	
		MCD BR	15.7	14.8	12.4	11.3	11.5	10.5	8.9	8.1	8.6	9.5	13.1	15.5	(38)	
(39)		MCWBR	5.8	5.4	5.0	4.6	3.9	3.3	3.0	3.1	3.6	4.0	5.4	6.1	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.526	0.558	0.679	0.701	0.725	0.714	0.638	0.588	0.573	0.589	0.528	0.501	(40)	
taud		1.848	1.756	1.532	1.561	1.530	1.590	1.762	1.873	1.894	1.826	1.859	1.907	(41)		
Ebn at Noon		766	763	686	672	645	644	698	742	755	729	758	775	(42)		
Edh at Noon		200	229	293	284	288	268	227	205	201	210	196	184	(43)		
(44)	All-Sky Solar Radiation	RadAvg	5.37	5.99	6.45	6.63	6.47	6.08	6.01	5.82	6.12	6.02	5.63	5.27	(44)	
(45)		RadStd	0.32	0.37	0.34	0.35	0.26	0.18	0.21	0.28	0.24	0.14	0.16	0.20	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page